

Olympiad Test Paper — Science (Class 7) — Paper 5

Chapter 19: Earth, Moon and the Sun

Paper 5 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	19 — Earth, Moon and the Sun
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. the hardest traps tempt you to use Earth's shadow for ordinary phases, to read 'crescent/gibbous' off the wrong orbit position, to confuse waxing-left with waning, to forget that the same near-side fact comes from one rotation per orbit, and to mix up apparent size (angular) with real size. Do not write on this paper — mark answers on your answer sheet.

1. An observer watches the Moon for one whole month and lists, in order, the fraction of the disc that looks lit each night: 0, then a thin sliver, then a half, then a fat shape, then the full disc, then back down through fat, half, sliver to 0 again. The single physical quantity that changes smoothly to produce this entire list is:

- (A) the fraction of the Moon that the Sun is lighting at any moment, which itself grows and shrinks
- (B) the size of Earth's shadow creeping across the Moon and then sliding off again
- (C) the distance between Earth and the Moon, which is largest at full Moon and smallest at new Moon
- (D) the angle between the directions Earth-to-Sun and Earth-to-Moon, which sets how much of the sunlit half faces us

2. At a certain instant the Moon sits so that the line from Earth to the Moon is at right angles to the line from Earth to the Sun, and the lit edge we see is on the right. One week earlier this same Moon would have appeared as:

- (A) a thin waning crescent lit on the left, low in the east before sunrise
- (B) a thin waxing crescent lit on the right, low in the west after sunset

- (C) a waxing gibbous, more than half lit, high in the south at sunset
- (D) a full Moon rising in the east as the Sun set in the west

3. A student draws the Sun far to the left, Earth in the middle, and asks where to place the Moon so that we on Earth see a full disc lit. The Moon must go:

- (A) on the far side of Earth from the Sun, so its whole sunlit face turns toward us
- (B) between Earth and the Sun, so the side facing us catches the most light
- (C) directly above Earth at a right angle to the Sun, so half its face is lit
- (D) anywhere, since the full Moon depends on the season, not on position

4. Over four successive nights an observer records the lit fraction as 0.30, 0.42, 0.55, 0.68, with the bright edge always on the right. The next night's most likely reading and name are:

- (A) about 0.20, a waning crescent, because the lit part is shrinking
- (B) about 0.55, a first quarter, because it stays half lit for several nights
- (C) about 0.80, a waxing gibbous, because the lit part is still growing past half
- (D) about 0.50, a last quarter, because right-lit Moons are always quarters

5. Half the Moon is in sunlight right now, yet tonight it appears as a slim crescent. A classmate insists this is impossible because 'half lit should look half lit.' The correct resolution is:

- (A) only a thin strip of the Moon is actually receiving sunlight at the moment
- (B) the lit half is mostly turned away from us, so only a thin edge of it shows from Earth
- (C) Earth's shadow is covering most of the lit half tonight
- (D) the Moon's own dark rock is absorbing most of the sunlight that lands on it

6. A diagram shows the Moon one-eighth of the way around its orbit from new Moon (an eighth of a full circle). The shape we would see is best described as:

- (A) a waxing gibbous, with more than half lit and growing
- (B) a first quarter, exactly half lit
- (C) a waxing crescent, with less than half the disc lit and growing
- (D) a full Moon, the whole disc lit

7. Two observers, one in India and one in Australia, both look at the same first-quarter Moon on the same night. The lit fraction each sees is:

- (A) different, because the two are on opposite sides of Earth and see opposite halves
- (B) the same, about one half, because the Sun-Earth-Moon angle is the same for both
- (C) fuller for the Australian observer, since the Southern Hemisphere is tilted toward the Moon
- (D) zero for one of them, because only the day side of Earth can see the Moon's lit face

8. If, hypothetically, the Moon's orbit were tilted so steeply that it passed almost over Earth's poles instead of near the equator plane of its motion, but it still went new-quarter-

full-quarter each orbit, the basic cause of the phases would:

- (A) stay the same, since phases come from the changing Sun-Earth-Moon angle regardless of orbit tilt
- (B) disappear, because phases need the Moon to stay in Earth's equatorial plane
- (C) reverse, so the Moon would wane while approaching the Sun
- (D) depend on Earth's seasons rather than on the Moon's position

9. A picture shows the lit portion as a 'D' shape, with the straight edge on the left and the bulging curved bright edge on the right, the lit part clearly bigger than half. This Moon is:

- (A) a waning gibbous, between full and last quarter
- (B) a waxing gibbous, between first quarter and full
- (C) a first quarter, exactly half lit
- (D) a waning crescent just before new Moon

10. On the same calendar night, a full Moon is high in the sky around midnight. Half an orbit (about two weeks) later, at the same midnight hour, the Moon will most likely be:

- (A) another full Moon, high overhead again at midnight
- (B) a first-quarter Moon setting in the west at midnight
- (C) a waning gibbous rising in the east at midnight
- (D) not visible, because it will be a new Moon, up in the daytime sky with the Sun

11. An astronomy puzzle asks: as the Moon moves from new toward full, the lit fraction we see goes 0 to 1; from full back to new it goes 1 to 0. The single word that correctly labels the first half of this journey is:

- (A) waning, meaning the lit part is shrinking night after night
- (B) gibbous, meaning the disc stays more than half lit throughout
- (C) eclipsing, meaning Earth's shadow is sliding off the disc
- (D) waxing, meaning the lit part is growing night after night

12. Tonight the Moon is lit on its left side and the lit part is smaller than last night's. Following this trend, in a few nights the Moon will:

- (A) grow to a right-lit first quarter and then to full
- (B) stay a left-lit half for about a week before changing
- (C) swing its lit side over to the right and begin waxing again immediately
- (D) shrink to a thin left-lit crescent and then reach new Moon

13. A waning crescent and a waxing crescent are photographed; both are slivers. Without any time stamp, the surest single clue to tell them apart in the Northern Hemisphere is:

- (A) which one is brighter, since the waxing crescent is always brighter
- (B) which side is lit: the waxing crescent is lit on the right, the waning on the left
- (C) which one is larger, since the waning crescent is always a bigger sliver
- (D) the colour, since the waning crescent is reddish and the waxing crescent is white

14. At full Moon the Moon rises in the east just as the Sun sets in the west, and they sit opposite each other in the sky. From this single geometric fact, at local midnight the full Moon will be:

- (A) just rising in the east, because it rose at sunset
- (B) high in the sky, near its highest point, opposite where the Sun is below the horizon
- (C) already setting in the west, because full Moons set early
- (D) below the horizon, because the Sun pulls it down at night

15. A waxing crescent Moon is seen only briefly, low in the west just after sunset, while a full Moon stays up all night. The reason the thin waxing crescent appears for so short a time is that it lies:

- (A) close to the Sun's direction in the sky, so it sets soon after the Sun does
- (B) opposite the Sun, so it follows the Sun down within minutes
- (C) directly overhead at sunset, so it falls quickly to the horizon
- (D) behind Earth's shadow, which blocks it for most of the night

16. An almanac says a certain Moon 'rises around midnight and is highest at dawn.'

Matching this rise-set timing to a phase, this Moon is most likely a:

- (A) first-quarter Moon, which rises around noon and is high at sunset
- (B) last-quarter (waning half) Moon, which rises near midnight and is high at sunrise
- (C) full Moon, which rises at sunset and is high at midnight
- (D) new Moon, which rises and sets with the Sun

17. An observer notices that, night after night for a week, the Moon rises later and shows more of its disc lit on the right. The single statement consistent with both observations is that the Moon is:

- (A) waxing and moving eastward in its orbit, so it lags farther behind the Sun each night while turning more of its lit half toward us
- (B) waning while drifting westward, so it rises earlier and shows less light
- (C) staying fixed in the sky while only its brightness changes
- (D) approaching the Sun, so it should be shrinking, not growing

18. Two students disagree: one says 'the lit edge of a crescent always points toward where the Sun is,' the other says 'it points away from the Sun.' Using the fact that the Moon shines by reflecting sunlight, the correct rule is:

- (A) the bright edge points away from the Sun, into the dark sky
- (B) the bright edge always points straight down toward the horizon
- (C) the bright edge points toward Earth's shadow
- (D) the bright curved edge points toward the Sun, since that side is the one catching sunlight

19. Just after a new Moon, the very thin crescent that appears in the west at dusk grows fatter each evening. Connecting 'grows fatter' to the orbit, this happens because the Moon

is:

- (A) moving closer to the Sun, so it receives more sunlight overall
- (B) entering a brighter region of space away from Earth's shadow
- (C) moving away from the line toward the Sun, so a larger slice of its sunlit half turns to face Earth
- (D) rotating faster so more of its surface gets lit each night

20. At first quarter the Moon is highest in the sky around sunset; at last quarter it is highest around sunrise. The reason these two half-Moons peak at opposite times of day is that they sit:

- (A) on opposite sides of the new-Moon line, one a quarter-orbit ahead of the Sun and one a quarter-orbit behind
- (B) at the same orbital point but seen from opposite hemispheres of Earth
- (C) at full-Moon and new-Moon positions respectively
- (D) at the same place in the sky but in different seasons

21. A learner sketches a 'D'-shaped Moon (straight edge left, bulge right) and labels it 'waning gibbous.' The single correction needed is:

- (A) the shape drawn is actually a crescent, not a gibbous
- (B) gibbous Moons are always lit on the left, so the bulge is on the wrong side
- (C) the shape is a quarter Moon and should not be called gibbous at all
- (D) a right-lit gibbous is waxing, not waning, so it should be labelled waxing gibbous

22. The Sun never lights more than half of the Moon at once, yet over a month we see the lit area we observe swing from nothing to the whole disc. The cleanest way to state why a 'full' Moon looks fully lit is:

- (A) at full Moon the entire sunlit half happens to be the half pointed at Earth
- (B) at full Moon the Sun briefly lights the whole sphere of the Moon
- (C) at full Moon Earth's light adds to the Sun's to fill in the dark half
- (D) at full Moon the Moon is closest to the Sun and so fully glows

23. We always see essentially the same markings on the Moon's near face. The reason, stated as a rate, is that the Moon:

- (A) does not rotate on its axis at all, so the same side is frozen toward us
- (B) rotates once a day, the same as Earth, keeping pace with our view
- (C) turns once on its axis in exactly the time it takes to orbit Earth once
- (D) rotates twice per orbit, which cancels out to show one face

24. Imagine the Moon kept orbiting Earth but its spin suddenly stopped completely. Over the next full orbit, an Earth observer would:

- (A) keep seeing exactly the same face, because spin has nothing to do with which side we see
- (B) see the Moon vanish, because only a spinning Moon reflects light to us

- (C) see the same face but upside down compared with before
- (D) gradually see all sides of the Moon, because a non-spinning Moon presents every face as it goes around

25. The hemisphere of the Moon that never faces Earth is often wrongly called the 'dark side.' During the lunar daytime there, that far side is actually:

- (A) permanently in darkness, which is why it is never seen from Earth
- (B) lit only by faint starlight, never by the Sun
- (C) brightly sunlit, since the far side gets just as much sunlight as the near side over a month
- (D) lit only when there is a full Moon as seen from Earth

26. A spacecraft can photograph the Moon's far side, but no person standing anywhere on Earth ever sees it directly. The single reason is that the Moon's:

- (A) far side is wrapped in permanent shadow that cameras can pierce but eyes cannot
- (B) rotation is synchronised with its orbit, always keeping the same hemisphere toward Earth
- (C) orbit is so fast that the far side blurs out of view from the ground
- (D) far side faces away from the Sun, so no light reaches Earth from it

27. On the Moon's near side, an astronaut would see Earth hanging in roughly the same patch of sky day after day, never rising or setting much. This curious fact follows from the Moon's:

- (A) lack of an atmosphere, which holds Earth steady in the lunar sky
- (B) fast rotation, which sweeps Earth around so quickly it looks fixed
- (C) synchronous rotation, which keeps the near side, and thus its view of Earth, fixed in one direction
- (D) great distance from Earth, which makes Earth appear motionless

28. A student claims that because we see slightly more than half the Moon's surface over time, the Moon must show us 'a different face every night.' The best correction is:

- (A) the Moon truly shows a completely different face each night as it tumbles freely
- (B) the Moon shows one face only because Earth's gravity glues it in place rigidly
- (C) the Moon shows different faces only during eclipses
- (D) the Moon shows essentially one face because it spins once per orbit; we glimpse only a little extra around the edges

29. If the Moon orbited Earth in the same time but rotated three times for every one orbit, an Earth observer would, over a single orbit:

- (A) still see only one face, because orbit time is all that matters
- (B) see the near side flicker between two halves only
- (C) see a full Moon three times as often
- (D) see the Moon's whole surface cycle past, not just one fixed face

30. The same-face fact and the long lunar day are two sides of one coin. A single sentence linking them is:

- (A) because the Moon has no atmosphere, it both keeps one face and has long days
- (B) because the Moon is small, it spins slowly and shows one face only by coincidence
- (C) because Earth is between the Moon and the Sun, the Moon's day is long
- (D) because the Moon spins once per orbit, it both keeps one face to Earth and takes about a month between its own sunrises

31. An astronaut on the far side of the Moon cannot directly radio Earth without a relay satellite. The geometric reason is that:

- (A) radio waves cannot travel through the Moon's atmosphere on the far side
- (B) the bulk of the Moon always sits between the far side and Earth because the same near side faces us
- (C) the far side is always in darkness, which blocks radio signals
- (D) Earth's rotation carries it out of view of the far side for half of each day

32. Day and night on Earth and the changing position of the Sun across the sky are both explained by one motion. To distinguish it from the motion that causes phases of the Moon, day and night are caused by:

- (A) Earth's orbit around the Sun, while the Moon's phases come from Earth's spin
- (B) the Moon's orbit around Earth, while phases come from Earth's orbit
- (C) Earth's spin on its own axis, while the Moon's phases come from the Moon's orbit around Earth
- (D) the Sun's orbit around Earth, while phases come from the Moon's spin

33. It is mid-morning in India while it is the middle of the night in a country on the opposite side of Earth. The single statement that explains both at once is:

- (A) the Sun has switched off over the far country while shining on India
- (B) Earth is a sphere lit on one side by the Sun, and its spin places the two countries on the lit and unlit sides at that instant
- (C) India is tilted toward the Sun this season while the far country is tilted away
- (D) the Moon is blocking sunlight from reaching the far country

34. An observer at a fixed spot watches the Sun, the stars, and the Moon all drift across the sky from east to west over the course of a night and day. The common cause of all three apparent motions is that:

- (A) Earth itself turns from west to east, so everything in the sky appears to slide the other way
- (B) the Sun, stars, and Moon all genuinely circle Earth once a day
- (C) Earth's yearly orbit drags the sky westward each night
- (D) the Moon's gravity pulls the whole sky along with it

- 35.** One sunrise-to-next-sunrise on the Moon takes about a month, far longer than Earth's 24 hours. The direct reason is that the Moon:
- (A) does not rotate, so the Sun simply hangs in one place forever
 - (B) rotates once every 24 hours like Earth but its days feel longer with no clocks
 - (C) orbits the Sun directly, which sets its long day length
 - (D) rotates on its axis just once per orbit, so the Sun returns to the same place in its sky only about once a month
- 36.** If Earth spun twice as fast as it does now, with everything else unchanged, the most direct effect would be:
- (A) a year lasting only half as long
 - (B) the Moon showing two new phases each month
 - (C) a day-night cycle of about twelve hours instead of twenty-four
 - (D) the seasons reversing between hemispheres
- 37.** On the equator the Sun climbs nearly straight up and sets quickly, giving very short twilight, while on the Moon there is essentially no twilight at all. The reason the Moon has almost no twilight is that it:
- (A) spins so slowly that the Sun jumps below the horizon instantly
 - (B) has no atmosphere to scatter sunlight, so the sky goes from lit to black abruptly
 - (C) is so small that the horizon is very close and the Sun drops fast
 - (D) is too far from the Sun for any twilight glow to form
- 38.** On the same night, an Indian observer and a Japanese observer (well to India's east) both see the full Moon. The Japanese observer sees it rise:
- (A) later, because eastern places always lag behind in the day
 - (B) at exactly the same clock instant, since the Moon is full for everyone at once
 - (C) only at midnight, regardless of where they stand
 - (D) earlier by clock time, because Earth's eastward spin brings eastern places into the Moon's view sooner
- 39.** We see the Moon at night because it reflects sunlight. If the Sun were somehow blocked from lighting the Moon for one night, the Moon would:
- (A) become invisible from Earth, since it has no light of its own to give off
 - (B) still glow faintly from heat stored in its rocks during the day
 - (C) shine with its own pale light, only dimmer than usual
 - (D) turn red, the way it does during a lunar eclipse
- 40.** A child argues, 'The Moon must make its own light, like a small star, because it shines steadily all night.' The single sharpest flaw is that:
- (A) the Moon shines only because Earth's light bounces onto it
 - (B) the Moon shines because it is hot enough to glow on its own

(C) the Moon only reflects the Sun's light, whereas a star produces its own light by nuclear reactions

(D) the Moon shines only during full Moon, not all night

41. Two astronauts, one on the Moon's near side at lunar noon and one on the far side also at lunar noon, are asked who sees the Sun. The correct answer is:

(A) both see the Sun, because each one's 'noon' means the Sun is overhead at that spot regardless of which side faces Earth

(B) only the near-side astronaut, because the far side never gets sunlight

(C) only the far-side astronaut, because the near side is shaded by Earth

(D) neither, because the Moon makes its own light and has no real noon

42. The phrase 'the Moon shines' is loose. The most scientifically precise rewording is:

(A) the Moon reflects sunlight toward Earth, making it appear bright

(B) the Moon emits its own moonlight from glowing minerals

(C) the Moon glows because Earth's atmosphere lights it up

(D) the Moon burns gases on its surface to produce light

43. At full Moon the entire near side is brilliantly lit; at new Moon the near side is dark. Using the reflected-light idea, the dark new Moon means that:

(A) the Sun has stopped lighting the Moon for those nights

(B) the sunlit half of the Moon is turned away from us, leaving the unlit half facing Earth

(C) Earth's shadow is fully covering the Moon at new Moon

(D) the Moon has moved into a region where no sunlight reaches

44. Sunlight reaches both Earth and the Moon, yet Earth's daytime sky is blue while the Moon's daytime sky stays black. The reflected-and-scattered-light reason is that:

(A) the Moon is too far from the Sun for blue light to reach it

(B) the Moon reflects only red light, so its sky cannot look blue

(C) the Moon's grey rock absorbs all the blue light from the Sun

(D) Earth's atmosphere scatters sunlight to make a blue sky, but the Moon has no atmosphere to scatter it

45. Footprints left on the Moon in 1969 are expected to remain almost unchanged for millions of years. The single underlying cause behind their long survival is that the Moon has:

(A) very strong gravity that presses the prints permanently into the rock

(B) a fast spin that packs the dust hard around each print

(C) essentially no atmosphere, so there is no wind or rain to erode or cover them

(D) thick clouds that shield the surface from being disturbed

46. Daytime temperatures on the Moon can exceed boiling water while night temperatures fall far below freezing. The single best explanation for this enormous swing is that the Moon:

- (A) has no atmosphere to trap heat or even out temperatures between day and night
- (B) is much closer to the Sun by day and much farther by night
- (C) spins so fast that one side bakes while the other freezes
- (D) has oceans that boil in the day and freeze at night

47. There is complete silence on the Moon: a struck rock makes no sound an astronaut a metre away can hear. The reason is that the Moon has:

- (A) gravity too weak to let sound waves move
- (B) no air, and sound needs a medium such as air to travel through
- (C) rock too hard for sound to pass into the surroundings
- (D) a magnetic field that cancels sound waves

48. A meteoroid that would flare up as a 'shooting star' high in Earth's sky instead strikes the Moon and digs a crater. The reason it does not burn up before landing on the Moon is that the Moon has:

- (A) no atmosphere, so there is no air friction to heat and burn the incoming rock
- (B) a thicker atmosphere that slows the rock without heating it
- (C) stronger gravity that pulls the rock down too fast to burn
- (D) a magnetic shield that protects the rock from burning up

49. Astronauts dropped a hammer and a feather together on the Moon and both hit the ground at the same instant, unlike on Earth. This shows that on the Moon:

- (A) gravity is so weak that the feather and hammer weigh exactly the same
- (B) the feather is heavier than on Earth because the Moon's dust clings to it
- (C) there is no air resistance, so all objects fall together regardless of shape or weight
- (D) the hammer falls slower to match the feather's speed

50. The Moon's surface keeps its ancient craters sharp, while most of Earth's old craters have worn away. The decisive difference is that:

- (A) the Moon has no wind, rain, or running water to erode its craters, whereas Earth does
- (B) the Moon's craters are made of harder rock than Earth's
- (C) the Moon is struck far more often, so craters are constantly refreshed
- (D) Earth's stronger gravity flattens its craters over time

51. A person who can jump 0.5 m high on Earth can jump much higher on the Moon with the same effort. The reason is that the Moon's surface gravity is about:

- (A) six times Earth's, which somehow launches the body upward
- (B) the same as Earth's, but the lack of air lets the body rise farther

- (C) one-sixth of Earth's, so the same push lifts a body much higher
- (D) zero, so the astronaut would float away with any push

52. The Moon and the Sun look nearly the same size in our sky, yet the Sun is hundreds of times wider than the Moon. The reconciliation is that:

- (A) the Sun and Moon really are about the same true size
- (B) the Moon is closer to the Sun than to Earth, which enlarges its look
- (C) Earth's atmosphere magnifies the Moon until it matches the Sun
- (D) the Sun is hundreds of times farther away, and its greater distance offsets its greater size so both look about the same

53. Ranking actual physical sizes from largest to smallest, the correct order for the Sun, Earth, and the Moon is:

- (A) Earth, then Sun, then Moon
- (B) Sun, then Earth, then Moon
- (C) Sun, then Moon, then Earth
- (D) Moon, then Earth, then Sun

54. Listing the Moon, the Sun, and the next-nearest star beyond the Sun in order of increasing distance from Earth (nearest first) gives:

- (A) the Sun, then the Moon, then the next-nearest star
- (B) the Moon, then the next-nearest star, then the Sun
- (C) the Moon, then the Sun, then the next-nearest star
- (D) the next-nearest star, then the Sun, then the Moon

55. A scale model places Earth as a 1 cm ball. On the same scale, the Moon would be a smaller ball placed about 30 Earth-diameters away, and the Sun an enormous ball very much farther off. The lesson this model teaches is that:

- (A) the Moon nearly touches Earth, since it is our closest neighbour
- (B) the Sun is only a little farther than the Moon
- (C) Earth, Moon, and Sun are packed close together in a tight cluster
- (D) space is mostly empty: the bodies are tiny compared with the gaps between them

56. Both Earth's single natural Moon and human-made artificial satellites keep circling Earth instead of flying off into space. The shared reason is that:

- (A) Earth's gravity continually pulls them toward Earth, bending their paths into orbits
- (B) they are held up by the cushion of Earth's atmosphere at that height
- (C) the Sun's light pressure pushes them around Earth
- (D) their fast spin creates a force that keeps them aloft

57. Suppose the Moon were moved to twice its present distance from Earth, everything else unchanged. Watched over a month from Earth, the Moon would:

- (A) stop showing phases, because phases need a fixed distance
- (B) still go through its usual phases, but look smaller in the sky

- (C) shine brighter, because it would catch more sunlight farther out
- (D) change colour from white to red because of the extra distance

58. If the Sun were replaced by a star of the same brightness but placed twice as far away, the most direct change for Earth (ignoring orbit effects) would be that:

- (A) the star would look exactly the same, since brightness is unchanged
- (B) the star would look larger because distance magnifies stars
- (C) the new star would look smaller and dimmer in our sky than the Sun does now
- (D) the star would no longer be classed as a star

59. An observer in the Southern Hemisphere watches a waxing Moon grow night after night and reports the lit edge is on the left, opposite to what a Northern Hemisphere book shows. The best explanation is:

- (A) the Sun-Earth-Moon geometry is the same, but the Southern observer's view of the sky is flipped, so the same waxing Moon appears lit on the opposite side
- (B) the Moon waxes from the left in the south because it orbits the other way there
- (C) the Southern Hemisphere sees an earlier phase, so the lit side genuinely differs
- (D) the Moon's lit half actually changes sides depending on which hemisphere it is over

60. A learner says, 'The Sun is special because it is the biggest, brightest thing in the universe.' Within the Class 7 picture, the accurate correction is:

- (A) the Sun is the only star that produces its own light
- (B) the Sun is the largest star that exists, with none bigger
- (C) the Sun is an ordinary star that looks huge and bright mainly because it is so close to us
- (D) the Sun is special because it reflects light from the other stars

Solutions — Science (Class 7), Chapter 19: Earth, Moon and the Sun — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	D	21	D	31	B	41	A	51	C
2	B	12	D	22	A	32	C	42	A	52	D
3	A	13	B	23	C	33	B	43	B	53	B
4	C	14	B	24	D	34	A	44	D	54	C
5	B	15	A	25	C	35	D	45	C	55	D
6	C	16	B	26	B	36	C	46	A	56	A
7	B	17	A	27	C	37	B	47	B	57	B
8	A	18	D	28	D	38	D	48	A	58	C
9	B	19	C	29	D	39	A	49	C	59	A
10	D	20	A	30	D	40	C	50	A	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (D) Exactly one half of the Moon is sunlit at all times; what changes is how much of that lit half points toward us, and that depends entirely on the Sun-Earth-Moon angle as the Moon orbits. The 'fraction the Sun lights' never changes from one half, so that choice is false. Earth's shadow only touches the Moon during the rare lunar eclipse, not during the ordinary monthly cycle, so the shadow option is the classic misconception. The Earth-Moon distance varies only slightly and is not tied to phase, so it cannot drive the lit-fraction list.

2. (B) A right-angle Moon lit on the right is a first quarter; it is waxing and lies a quarter of the orbit past new Moon. One week before first quarter the Moon was near new and just emerging as a thin waxing crescent, lit on the right and visible briefly in the west after sunset. A left-lit waning crescent belongs to the days before the next new Moon, not before first quarter. A gibbous comes after first quarter, not before. Full Moon is half an orbit past new, far later than a week before first quarter.

- 3. (A)** We see a full Moon when the entire sunlit hemisphere faces Earth, which happens when the Moon is opposite the Sun, on the far side of Earth. Placing it between Earth and Sun gives new Moon, because the lit side then faces away from us. A right-angle position gives a half-lit quarter, not full. Phase depends on orbital position relative to the Sun, never on season.
- 4. (C)** The numbers rise each night, so the Moon is waxing; having just crossed 0.50 it becomes a gibbous moving toward full, so the next reading should be larger, around 0.80, a waxing gibbous. A falling fraction would be waning, which contradicts the rising data. A quarter is a single instant at 0.50, not a value the Moon lingers on. A right-lit Moon at 0.50 is a first quarter, not a last quarter, and the figures are already past 0.50 anyway.
- 5. (B)** A sphere in sunlight always has one half lit, but our view of that lit half changes with the Sun-Earth-Moon angle; near new Moon the lit half points away, so only a sliver shows as a crescent. The Sun is not lighting just a strip; it lights a whole hemisphere always. Earth's shadow is not involved in ordinary phases. The Moon's rock is dull but it still reflects enough to be seen; the crescent shape is geometric, not about absorption.
- 6. (C)** First quarter is at one-quarter of the orbit (a right angle), and full Moon is halfway round. An eighth of the way is partway from new toward first quarter, so less than half is lit and growing: a waxing crescent. Gibbous and full come later, past the quarter and half marks respectively, so they are wrong for an eighth of an orbit.
- 7. (B)** Phase is set by the Sun-Earth-Moon angle, which is essentially the same for everyone on Earth at one instant, so both see a half-lit Moon. They may see it tilted differently or upside-down relative to each other, but the lit fraction is the same. Earth's hemispheres do not change which half of the Moon is sunlit. Both night-side observers can see whatever phase is up; the lit fraction does not drop to zero by location.
- 8. (A)** Phases arise purely from how much of the sunlit half faces us as the Sun-Earth-Moon angle changes; the tilt of the orbit does not remove that angle, so phases continue normally. They do not require any particular plane. Waxing always accompanies the lit half turning toward us as the Moon moves away from the Sun direction, so nothing reverses. Seasons are an Earth-tilt effect, unrelated to lunar phase cause.
- 9. (B)** More than half lit means gibbous; lit on the right (a fattening right side) means waxing, so it lies between first quarter and full: a waxing gibbous. A waning gibbous would be lit on the left as it shrinks toward last quarter. A first quarter is exactly half, not more. A waning crescent is a thin left-lit sliver, far less than half.
- 10. (D)** Half an orbit past full Moon is new Moon, which sits roughly in the Sun's direction and is therefore up during the day, invisible at midnight. It would not be full again until a whole cycle (about a month) passes. First quarter is a quarter of an orbit away, not half. A waning gibbous comes only a few days after full, not two weeks later.

11. (D) Growing lit fraction (0 toward 1) is waxing. Waning is the opposite, shrinking phase, which is the second half. Gibbous names only the more-than-half part of the cycle, not the whole growing half. Eclipsing involves Earth's shadow and the rare eclipse, not the routine waxing of phases.

12. (D) Left-lit and shrinking is a waning Moon heading toward new; it continues to thin into a waning crescent and then new Moon. It will not grow toward full, since that is waxing and would be right-lit in the Northern Hemisphere. The lit fraction changes a little every night, so it does not hold steady for a week. The lit side does not abruptly flip; only after passing new Moon does the next cycle begin lighting the right side.

13. (B) The two crescents differ in which limb is lit because they sit on opposite sides of the new-Moon position: waxing is right-lit, waning is left-lit (Northern Hemisphere). They can be equally bright and equally thin, so brightness and size do not separate them. Colour is not a phase property; both are the same pale reflected white.

14. (B) Being opposite the Sun, the full Moon crosses the meridian (its highest point) at midnight, the moment opposite the Sun's noon. It rose at sunset, so by midnight it is high, not still rising. It does not set until around sunrise, so it is not setting at midnight. The Sun does not pull the Moon below the horizon; that idea has no physical basis here.

15. (A) A thin waxing crescent is angularly near the Sun, so it trails the Sun across the sky and sets shortly after sunset, giving a brief window. Being opposite the Sun describes a full Moon, which is up all night, the opposite case. It is low in the west, not overhead, at sunset. Earth's shadow is irrelevant to ordinary crescent visibility.

16. (B) A last-quarter Moon lags the Sun by about six hours behind in rising, so it comes up near midnight and is highest near dawn. A first quarter leads opposite that, rising near noon and highest at sunset. A full Moon rises at sunset and peaks at midnight. A new Moon rises and sets with the Sun, visible to no one at night.

17. (A) The Moon orbits eastward, so each night it rises about 50 minutes later; meanwhile, moving away from the new-Moon direction turns more of its right-lit half toward us, so it waxes. Waning with a westward drift would mean rising earlier and shrinking, the opposite of both clues. The Moon does not stay fixed; it shifts eastward against the stars. Moving away from the Sun's direction is what makes it grow, so 'approaching the Sun' contradicts the growing data.

18. (D) Because the Moon glows only where sunlight strikes it, the lit limb faces the Sun, even when the Sun is just below the horizon. Pointing away from the Sun would mean the lit side is in shadow, which is impossible. The lit edge can point in any direction depending on the Sun's position, not fixed toward the horizon. Earth's shadow plays no role in lighting a crescent.

19. (C) After new Moon the Moon swings away from the Sun's direction, increasing the Sun-Earth-Moon angle, so more of its lit half faces us and the crescent fattens. The total sunlight it receives does not change with this swing; one half is always lit. Earth's shadow is not involved. The Moon's spin rate is constant and does not change how much of its lit half we see.

20. (A) First quarter trails a quarter orbit one way and last quarter the other way from new Moon, so one leads the Sun across the sky and the other lags, putting their high points at sunset versus sunrise. They are not the same point seen differently; they are genuinely opposite quarter positions. Neither is at full or new. Season does not set the daily high-point timing of a phase.

21. (D) In the Northern Hemisphere the right-lit growing Moon is waxing, so a right-bulging gibbous is a waxing gibbous, not waning. A gibbous is more-than-half lit, which a 'D' bulge represents, so it is not a crescent. Gibbous Moons can be lit on either side depending on waxing or waning; they are not 'always left.' A more-than-half disc is gibbous by definition, not a quarter.

22. (A) Full Moon means the sunlit hemisphere is exactly the hemisphere facing us, so we see a complete disc. The Sun never lights the whole sphere, only ever a half. Earthshine is faint and cannot brighten the far half into view. Distance to the Sun barely changes and is unrelated to the full phase, which is an alignment effect.

23. (C) The same-face fact comes from one spin per orbit (synchronous rotation): the Moon turns just enough each part of its orbit to keep the same side toward Earth. If it did not rotate at all, it would show every side over one orbit. A once-a-day spin would not match its month-long orbit and would not lock the face. Two spins per orbit would let us see different sides, not one face.

24. (D) Without spin, a Moon dragged around its orbit keeps the same fixed orientation in space, so the side facing Earth changes continuously and we would see all of it over one orbit. The same-face view actually requires synchronous spin, so stopping the spin breaks it. Reflection does not depend on spin, so the Moon would not vanish. The change is which side faces us, not an upside-down version of one face.

25. (C) The far side receives sunlight on the same monthly schedule as the near side; it is simply turned away from Earth, so 'far' is right but 'dark' is wrong. It is not permanently dark. Sunlight, not starlight, lights it during its day. Its lighting depends on the Sun's position relative to it, not on the phase Earth observers happen to see.

26. (B) Synchronous rotation keeps one hemisphere toward Earth at all times, so the far side is geometrically hidden from any earthbound viewer while a spacecraft can fly around to it. The far side is sunlit in turn, not in permanent shadow. Orbital speed does not blur a

hemisphere out of view. The far side does receive sunlight; it is just never turned toward us.

27. (C) Because the Moon keeps one face toward Earth, an observer on that face always looks toward Earth in nearly the same sky direction, so Earth barely moves for them. Atmosphere does not pin Earth in the sky. The Moon rotates slowly (once a month), not fast. Distance does not freeze Earth's apparent position; the synchronous spin does.

28. (D) Synchronous rotation keeps essentially one hemisphere toward us; small wobbles let us peek a little past the edges, but the face is basically constant, not different nightly. The Moon does not tumble freely. Gravity caused the lock long ago but does not rigidly glue a frozen face; the Moon does rotate, just in step with its orbit. Eclipses do not reveal new faces.

29. (D) One face stays toward us only when spin equals orbit (one-to-one). Three spins per orbit breaks the match, so the side facing Earth keeps changing and the whole surface parades past. Orbit time alone does not fix the face; the spin-to-orbit ratio does. The view would cycle through all longitudes, not just two halves. Phase frequency (full Moons) is set by the Sun-Earth-Moon orbit, not by how fast the Moon spins on its axis.

30. (D) One slow spin per orbit means the same side faces Earth and also means the Sun takes about a month to come back to the same point in the lunar sky, so both facts share the one-spin-per-orbit cause. Lack of atmosphere explains temperature swings, not the spin rate. Small size does not by itself set the spin rate, and the face-locking is not coincidence but synchronous rotation. Earth's position has nothing to do with the length of the lunar day.

31. (B) Synchronous rotation keeps the near side toward Earth, so the Moon's solid body always blocks a straight line from the far side to Earth, requiring a relay. The Moon has essentially no atmosphere, and radio is not blocked by darkness anyway. Earth never comes into the far side's sky because the same hemisphere of the Moon always faces us, not because of Earth's daily spin.

32. (C) Earth's daily axial spin turns us into and out of sunlight, giving day and night, whereas the Moon's monthly orbit changes the Sun-Earth-Moon angle that produces phases. Earth's yearly orbit gives the year, not the day. The Moon's orbit does not cause Earth's day. The Sun does not orbit Earth; that is the old, incorrect picture.

33. (B) At any instant the Sun lights one hemisphere of the spherical Earth; the spin carries different places into and out of that lit half, so two opposite places are in day and night at once. The Sun does not switch off. Day and night come from the daily spin, not from seasonal tilt, which affects day length and warmth, not the basic fact that opposite sides differ. The Moon is not blocking sunlight from a whole country.

34. (A) Earth's west-to-east spin makes all sky objects appear to move east to west together; the daily motion is a reflection of our rotation. The objects do not really circle Earth daily. The yearly orbit causes a slow seasonal shift of constellations, not the daily east-to-west sweep. The Moon's gravity does not drag the sky.

35. (D) The Moon's single slow spin per orbit means the Sun crosses the lunar sky about once a month, making each lunar day roughly a month long. It does rotate, so the Sun does not hang fixed. It does not spin once per 24 hours; that is Earth. The Moon orbits Earth, not the Sun directly, and it is the spin rate that sets day length.

36. (C) The length of the day is set by how fast Earth spins, so doubling the spin halves the day to about twelve hours. The year depends on the orbit, not the spin, so it is unchanged. The Moon's phase cycle depends on its orbit, not Earth's spin. Seasons follow from the axial tilt, which is not altered by spin speed.

37. (B) Twilight on Earth is scattered sunlight in the air; with no atmosphere the Moon has nothing to scatter light, so darkness arrives the instant the Sun sets. The slow spin actually makes lunar sunsets take a long time, but the sky still darkens abruptly because there is no air glow. Horizon distance does not create or remove twilight. Distance from the Sun does not produce twilight; an atmosphere does.

38. (D) Earth turns west to east, so places farther east are carried under any sky object first; the eastern observer sees moonrise earlier by the clock. Eastern places are ahead, not behind, in the daily cycle. Being full at one instant does not make moonrise happen at one clock time everywhere; rise time depends on longitude. Moonrise for a full Moon is near sunset locally, not fixed at midnight.

39. (A) The Moon is not a light source; it only reflects the Sun's light, so cutting off sunlight makes it dark and invisible. Stored heat does not make rock glow with visible light. The Moon has no self-light to fall back on. The red colour in a lunar eclipse comes from sunlight bent through Earth's atmosphere, which is exactly the light being removed here, so no red would appear.

40. (C) Stars generate light internally by nuclear reactions; the Moon merely reflects sunlight, so it is not a star. Earthshine is a faint secondary effect, not the main reason the Moon is bright. The Moon is not hot enough to glow visibly. The Moon is visible at many phases, but the point is the source of light, not how long it appears.

41. (A) Lunar noon means the Sun is highest at that location, and both near and far sides take their turn in sunlight over a month, so each at their own noon sees the Sun. The far side is not perpetually dark. Earth does not regularly shade the near side; that would be an eclipse, a rare event. The Moon has real day and night driven by sunlight, since it shines only by reflection.

42. (A) Precisely, the Moon reflects the Sun's light to us; it does not generate light. There are no self-glowing minerals producing moonlight. Earth's atmosphere does not light the Moon. There is no burning of gases; the Moon has essentially no atmosphere and no combustion.

43. (B) At new Moon the Moon is between us and the Sun, so the lit half points away and the dark half faces us; the Moon reflects light, but not toward Earth. The Sun keeps lighting one half always. Earth's shadow is on the opposite side at new Moon and is not involved. Sunlight does reach the Moon; it is the orientation, not a lack of light, that makes the near side dark.

44. (D) A blue sky is sunlight scattered by air molecules; without an atmosphere the Moon's sky has nothing to scatter sunlight, so it stays black even with the Sun shining. Distance does not strip blue from sunlight. The colour of reflected ground light does not set the sky colour. Surface rock absorption does not explain the black sky; the missing atmosphere does.

45. (C) No atmosphere means no wind and no rain, so nothing erases the prints; that lack of air is the root cause. The Moon's gravity is weak, not strong, and gravity does not preserve prints. A slow spin, not a fast one, characterises the Moon, and spin does not pack dust. The Moon has no clouds, again because it has no atmosphere.

46. (A) With no atmosphere there is no blanket to retain heat, so sunlit ground roasts and shaded ground plunges in temperature. The Moon's distance from the Sun does not lurch between day and night. Its spin is slow, not fast, and the swing is about the missing air, not spin. The Moon has no oceans; it has no liquid water at all.

47. (B) Sound travels as vibrations through a medium; with no atmosphere there is nothing to carry the sound, so it is silent. Gravity strength does not govern whether sound can propagate. Hard rock would conduct sound, but the air gap to the listener carries none. Magnetic fields do not cancel sound.

48. (A) Shooting stars glow from air friction in Earth's atmosphere; the airless Moon offers no such friction, so the rock reaches the ground intact and cratered. The Moon does not have a thicker atmosphere; it has essentially none. Gravity is weak, and burning is about friction, not fall speed. There is no magnetic shield doing this.

49. (C) On Earth air resistance slows a feather; the airless Moon removes that, so both fall together. Weak gravity slows the fall but does not make their weights equal, and equal weight is not why they land together. The feather's mass does not change on the Moon. The hammer does not slow down to match; rather, the feather stops being held back by air.

50. (A) Erosion by wind and water on Earth wears craters away; the airless, waterless Moon has no such weathering, so craters persist. Rock hardness is not the key difference;

weathering is. More impacts would not by itself preserve old craters. Earth's gravity does not flatten craters; weathering and geology erase them.

51. (C) Lunar gravity is roughly one-sixth of Earth's, so a given leg push carries a jumper much higher. Six times Earth's gravity would make jumping far harder, not easier. Gravity is not the same as Earth's; if it were, the jump height would not change much. Gravity is weak but not zero, so the astronaut comes back down.

52. (D) Apparent (angular) size depends on real size divided by distance; the Sun is both far bigger and far farther, and these nearly cancel, so the two look similar. They are not truly the same size; the Sun is vastly larger. The Moon is much nearer Earth than the Sun, not nearer the Sun. The atmosphere does not magnify the Moon to match the Sun.

53. (B) The Sun is by far the largest, Earth is much smaller, and the Moon is smaller still, giving Sun-Earth-Moon. Earth is not larger than the Sun. The Moon is smaller than Earth, so it cannot rank above Earth. The Moon is the smallest, not the largest, so the reversed order is wrong.

54. (C) The Moon is our closest neighbour, the Sun is much farther, and the next-nearest star is vastly farther than the Sun, giving Moon-Sun-star. The Sun is not closer than the Moon. The next-nearest star is far beyond the Sun, not between Moon and Sun. The star is the most distant, not the nearest.

55. (D) Scale models show large empty gaps relative to body sizes, teaching that space is mostly empty. Thirty Earth-diameters is a wide gap, so the Moon does not nearly touch Earth. The Sun is enormously farther than the Moon, not a little. The three are spread far apart, not packed tightly.

56. (A) Gravity from Earth bends a moving satellite's path into a closed orbit; this applies equally to the Moon and to artificial satellites. The atmosphere is far too thin at orbital heights to hold anything up. Sunlight pressure is far too weak to drive an orbit. A satellite's own spin does not keep it aloft; gravity and orbital motion do.

57. (B) Phases depend on the Sun-Earth-Moon angle, which still cycles as the Moon orbits, so phases continue; only the angular size shrinks with greater distance. Phases do not require a particular distance. Moving farther does not let the Moon catch more sunlight; it would look fainter and smaller. Distance does not turn its reflected white light red.

58. (C) Greater distance reduces both the apparent size and the apparent brightness reaching us, so the star looks smaller and dimmer than the present Sun. Equal intrinsic brightness does not keep apparent brightness the same once distance doubles. Distance shrinks, not magnifies, the apparent size. Moving a star farther does not change what kind of object it is.

59. (A) Phase and which physical half is lit are the same worldwide at one instant; a Southern observer simply views the sky inverted relative to a Northern one, so the same waxing Moon looks lit on the opposite limb. The Moon does not orbit a different way for the south. Both hemispheres see the same phase at the same time, not different phases. The Moon's lit half is fixed by the Sun's direction, not by which hemisphere is below it.

60. (C) The Sun is a fairly average star; it dominates our sky only because it is by far the closest. Many stars produce their own light. There are stars far larger than the Sun. The Sun makes its own light by nuclear fusion and does not merely reflect other stars' light.